BG 1st SEMESTER (NEP)
APPLIED COMPUTING: INTERNET BASICS & MULTIMEDIA COMPUTING (MINOR)

Session: December 2022 No. of Questions: 16 No. of Pages: 01
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 40

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. What is HTML?
2. Name any two authoring metaphors.
3. What is dithering?
4. What is the number of bits used in a grey-scale image?
5. What is lossless compression?
6. Name two lossy image compression techniques.
7. What is a motion vector?
8. What is YCbCr?

Section B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

9. Discuss the relation between multimedia and hypermedia.

OR

Briefly explain different elements of multimedia.

10. Explain GIF file format for image files.

OR

Differentiate between 8-bit grey-level and 8-bit colour images.

11. What is Variable Length Coding? Explain with an example.

OR

Explain the basic information theory.

12. Differentiate between MPEG-I and MPEG-II compression standards.

OR

Write the advantages of JPEG-LS standard over JPEG.

Section C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

13. Explain in detail any two authoring metaphors. Give examples in each.
14. Explain why we need to be able to have less than 24-bit colour and why this makes for a problem. Generally, what do we need to do to adaptively transform 24-bit colour values to 8-bit ones?
15. What is Distortion Measure? Explain Rate-Distortion Theory used in lossy compression.
16. How does JPEG compression technique work? Explain the different steps involved.